

Link Budget Report

Summary

#	Component	Signal Power Out (W)	Noise Power Out (W)	Signal Gain (dB)	Noise Gain (dB)	SNR (dB)
1	Signal source Signal starting at transmit side	1.0	0.0	inf	inf	inf
2	Path loss Path from transmitter to receiver	2.276573e-11	0.0	-1.0642e+02	inf	inf
3	Rx antenna gain Gain of the receive antenna	4.542361e-11	0.0	3.000000e+00	inf	inf
4	Noise source Noise at receive antenna	4.542361e-11	1.0	0.0	inf	-1.0342e+02
5	Quantization noise	4.542361e-11	1.000203e+00	0.0	8.834845e-04	-1.0342e+02
6	Sub-band Tuner	2.271181e-11	1.000203e-01	-3.0103e+00	-10.0	-9.6438e+01

Detailed Report

1. Signal source

name	Signal source
description	Signal starting at transmit side
signal power in (W)	0.0
noise power in (W)	0.0
signal power out (W)	1.0
noise power out (W)	0.0
snr	inf
noise gain	inf
signal gain	inf

2. Path loss

name	Path loss
description	Path from transmitter to receiver
signal power in (W)	1.0
noise power in (W)	0.0
signal power out (W)	2.2765734628573803e-11
noise power out (W)	0.0
speed of light	299792458.0
speed of light units	meters per second
free space path loss formula	$(4.0 * \text{Pi} * \text{frequency} * \text{distance} / \text{speed_of_light})^2$
snr	inf
noise gain	inf
signal gain	2.2765734628573803e-11

3. Rx antenna gain

name	Rx antenna gain
description	Gain of the receive antenna
signal power in (W)	2.2765734628573803e-11
noise power in (W)	0.0
signal power out (W)	4.542361237697535e-11

noise power out (W)	0.0
snr	inf
noise gain	inf
signal gain	1.9952623149688795

4. Noise source

name	Noise source
description	Noise at receive antenna
signal power in (W)	4.542361237697535e-11
noise power in (W)	0.0
signal power out (W)	4.542361237697535e-11
noise power out (W)	1.0
snr	4.542361237697535e-11
noise gain	inf
signal gain	1.0

5. Quantization noise

name	Quantization noise
description	
signal power in (W)	4.542361237697535e-11
noise power in (W)	1.0
signal power out (W)	4.542361237697535e-11
noise power out (W)	1.0002034505208426
quantization noise (W)	0.00020345052084257468
total bits	12.0
effective bits	8.678071905112638
headroom db	10.0
headroom bits	3.321928094887362
snr	4.5414372799175616e-11
noise gain	1.0002034505208426
signal gain	1.0

6. Sub-band Tuner

name	Sub-band Tuner
description	
signal power in (W)	4.542361237697535e-11
noise power in (W)	1.0002034505208426
signal power out (W)	2.2711806188487676e-11
noise power out (W)	0.10002034505208426
input lower frequency	10000000.0
input upper frequency	20000000.0
output lower frequency	15500000.0
output upper frequency	16500000.0
signal lower frequency in	15000000.0
signal upper frequency in	17000000.0
snr	2.270718639958781e-10
noise gain	0.1
signal gain	0.5